

Phase 1

Locus Characterization and Structural Definition of the 17p13.3 Microdeletion

1. Research Objective

Phase 1 establishes the structural genomic foundation for the larger investigation of an intersti-

tial 17p13.3 microdeletion and its possible downstream molecular, biochemical, neurological, and visual-system effects. The index deletion was reported as:

GRCh37/hg19 chr17:1,411,408–1,841,103

The objectives of this phase were to determine:

- 1. the exact genomic interval being studied;**
- 2. all identifiable genes and other annotated loci intersected by that interval;**
- 3. which protein-coding genes are fully versus partially deleted;**
- 4. whether the two breakpoint-intersected genes, INPP5K and RTN4RL1, lose actual protein-**
- coding sequence;
- 5. the relationship of each breakpoint to the current clinically standardized transcript;**
- 6. the approximate percentage of coding sequence removed from each breakpoint gene;**
- 7. which noncoding RNAs and pseudogenes are contained within the interval;**
- 8. which nearby loci should explicitly be excluded;**
- 9. and which structural questions remain unresolved because the reported clinical coordinates may**

not represent nucleotide-level chromosome junctions. This phase is deliberately restricted to structural genomic characterization. No visual, neurological, developmental, or other

phenotype is inferred from the presence of a gene inside the deletion. The final product of the phase is designated:

17p13.3 Deletion Map v1.0

2. Reproducibility Standard

A researcher reproducing this analysis should maintain a record of:

- database name;
- database or annotation release when available;
- access date;
- genome assembly;
- chromosome;
- genomic coordinates;
- transcript accession;
- transcript version;
- protein accession;
- strand;
- source URL or database accession;
- calculations performed;
- interpretation;
- and unresolved discrepancies.

The project should preferably use a directory structure such as:

```
Phase_1_Locus_Characterization/ | |——01_raw_coordinates/ |——02_gene_annotations/
|——03_transcripts/ |——04_clinvar_landmarks/ |——05_ccds/ |——06_non_coding_loci/
```

|—07_calculations/ |—08_figures/ |—09_source_archive/ |—README.txt A master

spreadsheet should contain at minimum:

Field

Description

Source

NCBI, Ensembl, ClinGen, etc.

Assembly

GRCh37, GRCh38

Chromosome

chr17

Start

Genomic start

End

Genomic end

Strand

+ or -

Gene

HGNC gene symbol

Gene type

Protein coding, ncRNA, pseudogene

Transcript

Accession.version

CNV state

Full / partial / outside

Coding affected

Yes / No / unresolved

Evidence

Source supporting classification

Access date

Date data were retrieved

Notes

Annotation discrepancies or caveats This prevents later phases from relying upon browser screenshots or undocumented manual interpretation.

3. Coordinate System and Genome Assembly

3.1 Freeze the genome assembly

The first requirement is to preserve the genome assembly exactly as reported. The index finding uses:

GRCh37 / hg19

For NCBI, the corresponding reference assembly is:

GRCh37.p13 — GCF_000001405.25

NCBI maintains GRCh37.p13 annotation resources, while Ensembl maintains a dedicated GRCh37 archive. The Ensembl GRCh37 gene annotation is based on an older frozen gene build, whereas some variation and phenotype information continues to receive updates. Consequently, Ensembl GRCh37 and current RefSeq/MANE annotations should not be expected to contain identical transcript versions. (ncbi.nlm.nih.gov) The working rule for this study is therefore: Do not mix GRCh37 and GRCh38 coordinates within a single structural calculation. If a source

provides only GRCh38 coordinates, those coordinates must either be excluded from the GRCh37 calculation or deliberately converted using a documented assembly-conversion method.

3.2 Preserve the original clinical coordinates

Create a source record containing: Assembly: GRCh37/hg19 Chromosome: 17 Start: 1,411,408 End: 1,841,103 Variant: deletion If the coordinates are treated as 1-based inclusive, the nominal interval length is:

$$1,841,103 - 1,411,408 + 1 = 429,696 \text{ bp}$$

or approximately:

429.7 kb

The clinical description of >430 kb should be preserved as reported rather than silently changed, because clinical microarray boundaries may reflect probe resolution, reporting conventions, or approximate interval designation. For calculations in this project, the explicit genomic coordinates take precedence over the rounded verbal size.

4. Coordinate-Convention Control

Coordinate conventions are a potential source of one-base replication errors. NCBI GFF3 annotation uses 1-based coordinates, with the start and end positions representing genomic feature boundaries. (ncbi.nlm.nih.gov) BED-based formats used by UCSC and many bioinformatics programs use a 0-based, half-open convention. (genome.ucsc.edu) Therefore the 1-based inclusive interval: chr17:1,411,408-1,841,103 would be represented in BED format as: chr17

17p13.3_index_deletion The original clinical coordinates should never be overwritten by this conversion. Instead preserve both: Clinical/1-based: chr17:1411408-1841103 BED/0-based

half-open: chr17 1411407 1841103 A peer attempting to reproduce the analysis should verify which convention is expected by every software package before entering coordinates.

5. Initial Browser Verification

Step 1 — Ensembl GRCh37

Open the Ensembl GRCh37 Archive and select the human GRCh37 assembly. Search: 17:1411408-1841103 The GRCh37 archive supports direct browsing of genes, transcripts, noncoding RNAs, structural variants, and regulatory annotations on the older assembly. (grch37.ensembl.org) Record:

- every visible protein-coding locus;
- every noncoding locus;
- strand orientation;
- apparent breakpoint-intersected genes;
- transcript IDs;
- and any loci lying very close to either boundary.

Do not yet interpret phenotype relevance. The purpose is to produce an initial candidate locus list.

Step 2 — NCBI Gene / Genome Data Viewer

Repeat the interval inspection using NCBI resources. For each candidate locus:

1. search the official gene symbol in NCBI Gene;
2. locate the Genomic context section;
3. select or identify the GRCh37.p13 row;
4. record the genomic start and end;
5. record strand;
6. record exon count where shown;
7. record RefSeq transcript and protein accessions.

The same gene should be checked independently in at least two annotation resources when it lies close to a deletion boundary.

6. Classifying Gene-Level CNV Overlap

For each gene with genomic interval:

$[Gs, Ge]$ and the deletion:

$[Ds, De]$ use the following rules.

Full deletion

A gene is completely encompassed if:

$Gs \geq Ds$ and

$Ge \leq De$

Partial deletion

A gene is partially affected if the two intervals overlap but at least one gene boundary extends outside the deletion. Formally:

$\max(Gs, Ds) \leq \min(Ge, De)$ while the full-deletion condition is not satisfied.

Outside

A gene is outside the deletion if:

$$Ge < Ds$$

or

$$Gs > De$$

These rules should be applied to genomic coordinates before any functional interpretation.

7. Identification of the Two Boundary Genes

Using this interval method, two protein-coding genes intersect the deletion boundaries:

- INPP5K at the lower-coordinate boundary;
- RTN4RL1 at the upper-coordinate boundary.

The remaining confirmed internal protein-coding genes fall completely inside the deletion. The next analytical problem is therefore not merely whether INPP5K and RTN4RL1 overlap the deletion, but whether the overlap includes coding sequence.

8. Selecting the Reference Transcript

8.1 Why MANE Select is used

Transcript selection is necessary because one gene may have multiple isoforms with different exons and coding boundaries. The Matched Annotation from NCBI and EMBL-EBI project (MANE) provides a standardized transcript pair in which the RefSeq and Ensembl/GENCODE representatives are matched in exon structure and coding sequence. MANE Select is intended to provide one representative transcript per locus and is widely useful for clinical reporting and genomic research. (ncbi.nlm.nih.gov) For Phase 1, the rule is:

Use the current MANE Select transcript as the primary transcript-level reference

whenever one is available. Alternative transcripts should be preserved separately rather than substituted silently.

8.2 INPP5K reference transcript

Record: Gene: INPP5K RefSeq MANE Select: NM_016532.4 Ensembl MANE Select: ENST00000421807.7 Protein: NP_057616.2 CCDS: CCDS11004.1 Protein length: 448 aa The CCDS database confirms that NM_016532.4 and ENST00000421807.7 are accepted MANE Select members of CCDS11004.1 and encode a 448-amino-acid protein. (ncbi.nlm.nih.gov)

8.3 RTN4RL1 reference transcript

Record: Gene: RTN4RL1 RefSeq MANE Select: NM_178568.4 Ensembl MANE Select: ENST00000331238.7 Protein: NP_848663.1 CCDS: CCDS45569.1 Protein length: 441 aa NCBI identifies RTN4RL1 as a two-exon minus-strand gene and reports NM_178568.4 / NP_848663.1 as the validated RefSeq transcript/protein pair, with CCDS45569.1 as its consensus coding sequence. (ncbi.nlm.nih.gov)

9. INPP5K Breakpoint Mapping

9.1 Establish genomic orientation

NCBI places INPP5K on the minus strand. This is critical because transcript direction is opposite increasing chromosome coordinates. For a minus-strand gene: higher genomic coordinates = transcript 5' side lower genomic coordinates

= transcript 3' side The deletion begins at: 1,411,408 inside INPP5K. The deleted portion runs from that point toward the higher-coordinate end of the gene. Therefore: The index deletion removes the 5' portion of INPP5K.

10. Using ClinVar as Transcript Landmarks

ClinVar variant reports provide genomic locations, HGVS expressions, transcript information, protein changes, and assembly-specific mappings. ClinVar itself distinguishes calculated molecular consequence from experimentally demonstrated functional consequence,

which is important when interpreting these data. (ncbi.nlm.nih.gov) The method used here is not to infer pathogenicity from the variants. Instead, known variant coordinates are used as landmarks linking transcript position to GRCh37 genomic position.

10.1 Search procedure

For INPP5K:

1. open ClinVar;

2. search INPP5K NM_016532.4;

3. verify that the result uses transcript NM_016532.4;

4. locate the GRCh37 genomic mapping;

5. record both:

- HGVS cDNA coordinate;
- GRCh37 genomic coordinate;

6. compare that genomic coordinate with the deletion boundary.

The following landmarks were obtained:

Classification for structural Transcript position

GRCh37 position

mapping c.165

1,416,843

Deleted side c.277

1,413,078

Deleted side c.419

~1,412,607

Deleted side

~1,412,457

Deleted side c.554+16 c.793

~1,401,400

Retained side The scientific interpretation is independent of the clinical classification of those individual variants. The only question being asked is: Does this known transcript coordinate map inside or outside the deletion?

11. INPP5K Breakpoint Bracketing

Because: c.554+16 → deleted while: c.793 → retained the deletion breakpoint lies between those transcript regions. This establishes that the deletion extends beyond coding position c.554 but does not reach the region containing c.793. Direct CCDS/exon reconciliation further places approximately the first 666 coding nucleotides of NM_016532.4 on the deleted chromosome segment under the reported-coordinate model. Because the INPP5K protein contains 448 amino acids:

$$448 \times 3 = 1344$$

amino-acid-coding nucleotides. The fraction removed is therefore approximately:

$$1344 = 0.4955$$

or:

49.6%

Reproducible INPP5K result

Gene: INPP5K CNV state: Partial Strand: - Affected side: 5' Coding sequence affected: Yes Estimated amino-acid-coding sequence removed: ~49.6% Transcript: NM_016532.4 Model limitation: based on reported CNV coordinates

12. RTN4RL1 Breakpoint Mapping

12.1 Establish orientation

RTN4RL1 is also located on the minus strand. Its GRCh37 gene span is approximately: chr17:1,837,971-1,928,628 while the deletion ends at: 1,841,103 Therefore only the low-coordinate portion of the genomic gene span overlaps the deletion. Because the gene is on the minus strand, the low-coordinate segment corresponds to the 3' side of the transcript. Thus: The deletion removes the 3' portion of RTN4RL1.

13. Confirming RTN4RL1 Coding Loss

Repeat the ClinVar landmark procedure using: NM_178568.4 A useful landmark is: NM_178568.4:c.1056G>A p.Pro352= which maps to approximately: GRCh37 chr17:1,840,060 Because:

$$1,411,408 < 1,840,060 < 1,841,103$$

this coding position lies inside the deleted interval. Therefore: RTN4RL1 coding sequence is definitively intersected. This conclusion does not depend upon whether the variant itself is pathogenic, benign, or synonymous. The variant is simply being used as a coordinate marker.

14. RTN4RL1 Coding-Overlap Calculation

CCDS45569.1 contains the consensus coding sequence corresponding to the 441-amino-acid RTN4RL1 protein. Under the reported-coordinate model, approximately: 1,313 of 1,326 CCDS nucleotides fall on the deleted side of the breakpoint. The estimated CCDS loss is therefore:

$$1326 = 0.9902$$

or approximately:

99.0%

Reproducible RTN4RL1 result

Gene: RTN4RL1 CNV state: Partial Strand: - Affected side: 3' Coding sequence affected: Yes Estimated CCDS nucleotide sequence removed: ~99.0% Transcript: NM_178568.4 Model limitation: based on reported CNV coordinates This illustrates why the percentage of the entire genomic gene span deleted is not a reliable proxy for the fraction of coding sequence disrupted.

15. Direct GFF3 Validation

The browser and ClinVar procedure should be independently checked against an assembly-matched annotation file. NCBI provides GFF3 annotations for annotated genome assemblies, and GFF3 start/end positions are represented as 1-based genomic coordinates. (ncbi.nlm.nih.gov) For GRCh37, use an annotation corresponding to: Assembly: GRCh37.p13 Accession: GCF_000001405.25 The NCBI GRCh37 annotation report can be used to document the annotation release associated with the downloaded file. (ncbi.nlm.nih.gov)

15.1 Manual GFF3 procedure

Download the GRCh37.p13 GFF3 annotation and search for: NM_016532.4 and: NM_178568.4 Extract every row where feature type is: exon or: CDS Record:

- chromosome accession;
- feature type;
- start;
- end;
- strand;
- phase;
- transcript accession.

Create one table per transcript. Example:

Transcript

Feature

Start

End

Strand

Overlap with CNV

NM_016532.4

CDS

X

Y

–

Full/Partial/None

NM_016532.4

CDS

X

Y

–

Full/Partial/None The exact rows should be preserved in the project archive.

16. Computational CDS-Overlap Calculation

For each coding segment:

$[Cs, Ce]$ and deletion interval:

$[Ds, De]$ calculate:

$Overlap = \max(0, \min(Ce, De) - \max(Cs, Ds) + 1)$ Then sum across all CDS segments:

$$L_{deleted} = \sum$$

$$Overlap_i$$

i Calculate total coding length:

$$LCDS = \sum$$

$$(Ce_i - Cs_{i+1})$$

i Then:

$$\%CDS \text{ deleted} = L_{deleted}$$

$$\times 100$$

$LCDS$ If reporting amino-acid-coding sequence, exclude the stop codon from the denominator. If reporting CCDS nucleotide sequence, explicitly state whether the stop codon is included. This distinction explains why the INPP5K result is reported as approximately 49.6% of amino-acid-

coding sequence, whereas the RTN4RL1 result is described as approximately 99% of CCDS se-

quence.

17. Optional BEDTools Replication

A researcher using BEDTools may create the deletion BED file: chr17

index_deletion and convert the relevant CDS annotations into BED coordinates. Then

use: `bedtools intersect \ -a transcript_CDS.bed \ -b index_deletion.bed \ -wao` The output

provides the number of overlapping bases for each CDS feature. Important: GFF3 coordinates

must be converted correctly from 1-based inclusive to 0-based half-open BED coordinates before

performing this analysis. Failure to convert coordinate conventions consistently will produce boundary errors.

18. IGV Visual Quality-Control Procedure

IGV may be used as an independent visual check.

Procedure

1. open IGV;

2. select the hg19/GRCh37 human genome;

3. navigate to:

chr17:1,350,000-1,950,000

4. load the deletion as a BED track;

5. display gene/transcript annotation;

6. zoom into the lower breakpoint around:

chr17:1,411,408

7. zoom into the upper breakpoint around:

chr17:1,841,103

8. confirm visually that:

- INPP5K crosses the lower deletion boundary;
- RTN4RL1 crosses the upper deletion boundary;
- the internal genes lie within the deletion;
- RTN4RL1-AS1 lies outside the interval.

IGV should be treated as visual quality control, not as the sole source of numeric coordinates. Numeric calculations should derive from exported annotation data.

19. Census of Fully Deleted Protein-Coding Genes

After establishing the breakpoint genes, query every locus between the two boundaries.

For each candidate:

1. retrieve its GRCh37 gene coordinates;
2. apply the full/partial/outside equations;
3. independently confirm boundary-near genes;
4. record gene type;
5. avoid phenotype interpretation at this stage.

The final confirmed protein-coding inventory is:

Gene

Structural classification

INPP5K

Partial; 5' coding disruption

Fully encompassed

PITPNA

SLC43A2

Fully encompassed

SCARF1

Fully encompassed

RILP

Fully encompassed

PRPF8

Fully encompassed

TLCD2

Fully encompassed

Fully encompassed

WDR81**SERPINF2**

Fully encompassed

SERPINF1

Fully encompassed

SMYD4

Fully encompassed

RPA1

Fully encompassed

RTN4RL1

Partial; 3' coding disruption Thus:

11 fully encompassed protein-coding genes

2 partially coding-disrupted protein-coding genes

13 affected protein-coding loci total

20. Noncoding RNA Census

A protein-coding-only search is insufficient for complete structural characterization.

Repeat the interval query while including:

- lncRNA;
- antisense RNA;
- microRNA;

- host RNA genes;
- pseudogenes;
- other annotated RNA loci.

20.1 PITPNA-AS1

NCBI identifies PITPNA-AS1 as a validated antisense noncoding RNA locus. Relevant validated transcript: NR_028514.1 Coordinate comparison places the locus completely inside the deletion. Classification: FULL

20.2 MIR22HG

Record: Gene: MIR22HG GRCh37: chr17:1,614,798-1,619,504 Strand: - Type: noncoding RNA host gene Validated transcripts include: NR_028503.2 NR_028504.2 NR_028505.2 All genomic coordinates fall inside the deletion. Classification: FULL

20.3 MIR22

Record: Gene: MIR22 GRCh37: chr17:1,617,197-1,617,281 Strand: - Type: microRNA RefSeq: NR_029494.1 Classification: FULL No phenotype interpretation is assigned at this stage.

21. Pseudogene Census

An expanded interval search identified: RN7SL105P with GRCh37 coordinates: chr17:1,507,668-1,507,966 The locus lies completely inside the deletion. Classification: FULL Its biological relevance to the study is currently: UNSUPPORTED / UNKNOWN It should nevertheless be retained in the structural map because Phase 1 records what is physically deleted, not merely what is presently believed to be clinically significant.

22. Explicit Negative Boundary Check: RTN4RL1-AS1

Every boundary analysis should also document the nearest loci that are not deleted. NCBI places: RTN4RL1-AS1 at approximately: GRCh37 chr17:1,921,010-1,924,000 The index deletion ends at: 1,841,103 Therefore:

$$1,921,010 > 1,841,103$$

and RTN4RL1-AS1 lies approximately 80 kb beyond the index interval. Classification: OUTSIDE This negative result is important because larger 17p13.3 deletions can encompass RTN4RL1-AS1. A peer should therefore not copy gene lists from broader “17p13.3 deletion” reports without performing coordinate-level comparison.

23. Minimum Structural Inventory

After combining protein-coding, noncoding, and pseudogene results, the minimum confirmed affected locus inventory is:

Protein-coding

Partial 1. INPP5K 2. RTN4RL1

Full 3. PITPNA 4. SLC43A2 5. SCARF1 6. RILP 7. PRPF8 8. TLCD2 9. WDR81 10. SERPINF2 11. SERPINF1 12. SMYD4 13. RPA1

Noncoding

14. PITPNA-AS1

15. MIR22HG

16. MIR22

Pseudogene

17. RN7SL105P

Explicitly excluded

RTN4RL1-AS1 Therefore:

At least 17 structurally confirmed loci are affected by the index deletion under the

present annotation scope. The phrase at least is necessary. This inventory does not yet constitute a complete regulatory map.

24. Transcript-Version Quality Control

The project identified discrepancies between current MANE annotations and the historical Ensembl GRCh37 gene build. This is expected. Ensembl states that the GRCh37 gene annotation is based on its older frozen annotation rather than the continuously updated main human gene annotation. (grch37-mart.ensembl.org) For example:

INPP5K

Current: NM_016532.4 ENST00000421807.7 Historical GRCh37 archive representation:
NM_016532.3 ENST00000421807.2

RTN4RL1

Current: NM_178568.4 ENST00000331238.7 Historical GRCh37 archive representation:
NM_178568.2 ENST00000331238.6 MANE itself notes that MANE transcripts are defined against GRCh38, while mappings are made available for GRCh37 and other assemblies for

clinical interpretation. (ncbi.nlm.nih.gov) The replication rule is therefore: Record transcript accession and version exactly.

Do not assume that the tran-

script displayed by an older GRCh37 browser is the latest clinically standardized

transcript.

25. ClinGen Dosage Data: Record but Do Not Interpret Yet

Once the structural map is frozen, a researcher may record ClinGen dosage-sensitivity status for each protein-coding gene. ClinGen evaluates evidence for haploinsufficiency and triplosensitivity at gene and genomicregion levels. (clinicalgenome.org) For Phase 1, however, dosage information should be recorded as metadata only. Example fields:

Gene

HI score

TS score

Curation date

Evidence status Functional prioritization based on those scores belongs to Phase 2. This prevents Phase 1 from selecting genes merely because they fit an anticipated phenotype.

26. Breakpoint-Resolution Limitation

The strongest limitation in Phase 1 concerns the meaning of the clinical CNV endpoints. The calculations treat: chr17:1,411,408-1,841,103 as the operational deleted interval. If these values originated from chromosomal microarray analysis, they may represent the positions of the last informative probes or another assay-defined boundary rather than the exact DNA junction. Therefore the analysis establishes:

feature overlap at the resolution of the reported CNV

but not necessarily: the exact nucleotide at which the chromosome physically broke and rejoined. Consequently, statements such as: ~49.6% INPP5K coding loss ~99.0% RTN4RL1 CCDS loss must be qualified: under the reported-coordinate model. Nucleotide-resolution confirmation would require an assay capable of identifying the physical breakpoint, such as appropriate whole-genome sequencing, targeted breakpoint sequencing, long-read sequencing, or another validated breakpoint-resolution method.

27. Independent Peer-Replication Checklist

A peer should be able to reproduce Phase 1 without seeing the project's conclusions beforehand. The recommended blinded replication sequence is:

A. Freeze input

☐Record GRCh37 ☐Record chr17:1411408-1841103 ☐Record original clinical size description separately

B. Verify assembly

☐Confirm GRCh37.p13 / GCF_000001405.25 ☐Do not substitute GRCh38 coordinates

C. Census loci

☐Search Ensembl GRCh37 ☐Search NCBI ☐Create independent gene list ☐Include ncRNAs and pseudogenes ☐Record strand orientation

D. Classify overlap

☐Full ☐Partial ☐Outside

E. Resolve breakpoint transcripts

☐Identify MANE Select ☐Preserve accession.version ☐Record CCDS ☐Record protein length

F. Map INPP5K

- ☐ Confirm minus strand ☐ Verify ClinVar transcript landmarks ☐ Extract CDS coordinates
- ☐ Calculate CDS overlap ☐ Compare with ~49.6%

G. Map RTN4RL1

- ☐ Confirm minus strand ☐ Verify c.1056 landmark ☐ Extract CDS coordinates ☐ Calculate CDS overlap ☐ Compare with ~99%

H. Negative controls

- ☐ Confirm RTN4RL1-AS1 is outside ☐ Check other loci immediately outside both boundaries

I. Annotation reconciliation

- ☐ Compare current MANE against GRCh37 Ensembl ☐ Record version discrepancies rather than replacing them

J. Archive

- ☐ Save GFF/GTF files ☐ Save tables ☐ Record database release ☐ Record access date ☐ Save calculation sheet or script
- If the peer's result differs, the discrepancy should first be investigated for:

- 1. assembly mismatch;**
- 2. transcript-version mismatch;**
- 3. coordinate-convention mismatch;**
- 4. inclusive versus half-open interval treatment;**
- 5. alternative transcript selection;**
- 6. annotation-release differences.**

Only after these are excluded should the biological interpretation itself be questioned.

28. Replication Output Expected

A successful independent replication should approximately recover:

Feature

Expected result

Assembly

GRCh37/hg19

CNV

chr17:1,411,408–1,841,103

Nominal span

429,696 bp

INPP5K

Partial, 5' coding disruption

INPP5K coding loss

~49.6% under coordinate model

RTN4RL1

Partial, 3' coding disruption

RTN4RL1 CCDS loss

~99% under coordinate model

Full coding genes

Total affected coding genes

Confirmed ncRNA genes

Confirmed pseudogene

Minimum loci

RTN4RL1-AS1

Outside A replication does not need to reproduce every historical transcript ID identically if annotation databases have changed after the original analysis. It must instead document the annotation version used and explain any discrepancy.

29. Evidence Classification

Structurally Established

The following findings are supported at the resolution of the reported CNV:

- INPP5K is partially deleted.
- INPP5K protein-coding sequence is affected.
- The deleted INPP5K sequence lies on the transcript's 5' side.
- Approximately 49.6% of the MANE amino-acid-coding sequence lies within the operational deletion interval.
- RTN4RL1 is partially deleted.
- RTN4RL1 protein-coding sequence is affected.
- The deleted RTN4RL1 sequence lies on the transcript's 3' side.
- Approximately 99% of its CCDS nucleotide sequence lies within the operational deletion interval.
- Eleven internal protein-coding genes are completely encompassed.
- PITPNA-AS1, MIR22HG, and MIR22 are completely encompassed.
- RN7SL105P is encompassed.
- RTN4RL1-AS1 is outside the deletion.
- At least 17 annotated loci are structurally affected.

Structurally Unresolved

Phase 1 does not establish:

- nucleotide-exact chromosome junctions;
- whether INPP5K produces a truncated RNA;
- whether RTN4RL1 produces a truncated RNA;
- nonsense-mediated decay;
- transcript stability;
- altered protein production;
- promoter activity;
- enhancer effects;
- compensatory expression from the intact allele;
- tissue-specific dosage effects.

Phenotypically Unresolved

Phase 1 does not establish:

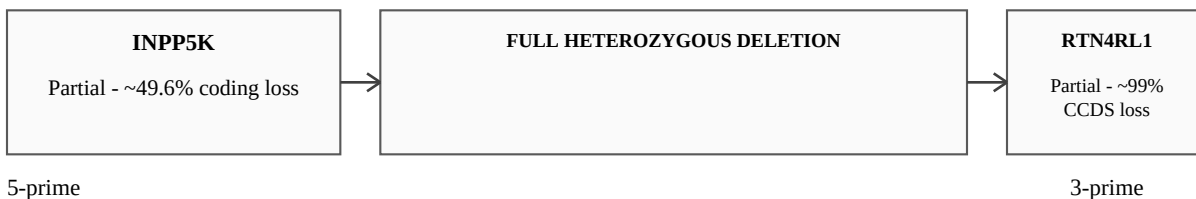
- visual stress;
- photophobia;
- magnocellular dysfunction;
- retinal dysfunction;
- cortical hyperexcitability;
- seizures;
- hearing loss;
- developmental delay;
- intellectual disability;

- congenital heart disease causality;
- or any other genotype–phenotype relationship.

These are deliberately deferred.

30. Phase 1 Final Structural Model

The deletion can now be represented as:



This diagram describes the affected chromosome only. The homologous chromosome is not represented and may retain intact copies unless additional variation is present.

31. Phase 1 Conclusion

Phase 1 establishes the index CNV as a structurally defined interstitial 17p13.3 deletion rather

than a generic 17p13.3 syndrome interval. The most important structural findings are that both breakpoint genes lose coding sequence and that their functional coding disruption is substantially different from what genomic-span overlap alone would imply. **INPP5K** loses approximately half of its MANE protein-coding sequence from the affected chromosome under the reported-coordinate model. **RTN4RL1** presents an even more pronounced discrepancy between genomic-span overlap and coding disruption: although the deletion intersects only a small proportion of the total genomic locus, it encompasses approximately 99% of the relevant CCDS nucleotide sequence. The internal interval completely encompasses at least eleven additional protein-coding genes, three established noncoding RNA genes, and one RNA pseudogene. The resulting minimum map contains:

17 affectedannotatedloci

No phenotype has been inferred from this finding. The purpose of Phase 1 is to establish a reproducible genomic substrate upon which later biochemical and functional analyses can operate.

32. Phase Status

PHASE 1 — COMPLETE

Frozen Structural Reference

Designation: 17p13.3 Deletion Map v1.0

Assembly: GRCh37/hg19

Interval: chr17:1,411,408–1,841,103

Future work must use this structural reference unless:

- 1. an amended clinical report changes the CNV coordinates;**
- 2. breakpoint-resolving sequencing identifies the physical junction;**
- 3. a demonstrable calculation error is identified;**
- 4. or a future annotation update materially changes transcript interpretation.**

If any of those occur, the reference should not be silently edited. It should instead become:

Deletion Map v1.1 or v2.0

with the change documented.